Aman Malik

Curriculum Vitae

aman.malik@posteo.net
https://amanmalik.net
4R4fYgwAAAAJ
aman-malik-181318234
0000-0002-7310-8448

Profile

Areas of expertise
- **Energy System Modelling:** Scenario building and net-zero pathways development using models like GCAM and REMIND.
- **Carbon Markets & Policy:** Modelling-based research on India's Carbon Credit Trading Scheme (CCTS) and emission trading frameworks.
- **Transition Risk:** Climate-related financial risk assessments for the financial sector and institutional actors.
- **Sub-national Policy:** Developing state-level energy models and net-zero strategies (e.g., Gujarat and Uttarakhand).

Interests: Climate change mitigation policy; Indian energy policy; modelling energy and climate policy scenarios; employment implications of mitigation; climate-related financial risk; carbon dioxide removal.

Education

10/2017–11/2021 **Dr. rer. nat.**, *Technical University of Berlin and Potsdam Institute for Climate Impact Research*, Berlin and Potsdam, Germany

Title: Barriers to power sector decarbonisation in India

10/2014–05/2017 **Master of Science in Integrated Climate System Sciences**, *University of Hamburg*, Hamburg, Germany
Specialised in climate economics and carbon dioxide removal

Thesis Title: Kinetics of olivine dissolution in column experiments

Grade: 1.24 (*excellent*)

2009–2013 **Bachelor of Technology in Chemical Engineering**, *Guru Gobind Singh Indraprastha University*, New Delhi, India
First Division, *(72.4%)*

Work Experience

06/2023–02/2026 **Programme Lead**, *Council on Energy, Environment and Water*, New Delhi, India
Low Carbon Economy Team

10/2022–05/2023 **Consultant**, *Council on Energy, Environment and Water*, New Delhi, India

07/2022–12/2024 **Guest Scientist**, *Potsdam Institute for Climate Impact Research*, Potsdam, Germany

01/2022–06/2022 **Postdoctoral Researcher**, *Potsdam Institute for Climate Impact Research*, Potsdam, Germany

10/2017–12/2021 **Researcher**, *Potsdam Institute for Climate Impact Research*, Potsdam, Germany
Group of Energy Systems, Research Department III - Transformation Pathways

06/2017–08/2017 **Consultant**, *Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)*, New Delhi, India

10/2016–12/2016 **Student Assistant**, *Biogeochemistry Laboratory, Institute of Geology, University of Hamburg*, Hamburg, Germany

05/2016–09/2016 **Student Assistant**, *Climate Science and Policy Team, Climate Analytics*, Berlin, Germany
Development of SIAMESE — a model to emulate Integrated Assessment Models

02/2015–03/2016 **Student Assistant**, *Biogeochemistry Laboratory, Institute of Geology, University of Hamburg*, Hamburg, Germany

Peer-reviewed Publications

2026 K. Dhandhania, **A. Malik**, R. Varma, V. Chaturvedi. "The Criticality of Country-Specific Scenarios to Inform Transition Risk Assessment for the Financial Sector: A case study for India". *Energy and Climate Change*. doi:10.1016/j.egycc.2026.100255

2025 P. Das and **A. Malik**. "Challenges to Deep-Decarbonization to Achieve Net Zero for India: A Review". *Strategic Analysis*.

2024 **A. Malik**, V. Chaturvedi, M. Sandhani, P. Das, C. Arora, N. Singh, R.Y. Cui, et al. "Implications of an emission trading scheme for India's net-zero strategy: a modelling-based assessment". *Environmental Research Letters*.

2022 T. Amann, J. Hartmann, R. Hellmann, E.T. Pedrosa, **A. Malik**. "Enhanced weathering potentials—the role of in situ CO_2 and grain size distribution". *Frontiers in Climate*.

2022 **A. Malik** and C. Bertram. "Solar energy as an early just transition opportunity for coal-bearing states in India". *Environmental Research Letters*.

2022 **A. Malik**. "Barriers to power sector decarbonisation in India". *PQDT Global* (Doctoral Dissertation).

2022 HL. van Soest... **A. Malik**, et al. "Global roll-out of comprehensive policy measures may aid in bridging emissions gap". *Nature Communications*.

2021 C. Bertram... **A. Malik**, et al. "Energy system developments and investments in the decisive decade for the Paris Agreement goals". *Environmental Research Letters*.

2021 L. Baumstark... **A. Malik**, et al. "REMIND2.1: Transformation and innovation dynamics of the energy–economic system within climate and sustainability limits". *Geoscientific Model Development*.

2021 **A. Malik**, C. Bertram, E. Kriegler, G. Luderer. "Climate policy accelerates structural changes in energy employment". *Energy Policy*.

2021 C. Bertram... **A. Malik**, et al. "COVID-19-induced low power demand and market forces starkly reduce CO_2 emissions". *Nature Climate Change*.

2020 **A. Malik**, C. Bertram, J. Després, J. Emmerling, S. Fujimori, A. Garg, et al. "Reducing stranded assets through early action in the Indian power sector". *Environmental Research Letters*.

2020 H. Wang, W. Chen, C. Bertram, **A. Malik**, E. Kriegler, G. Luderer, et al. "Early transformation of the Chinese power sector to avoid additional coal lock-in". *Environmental Research Letters*.

2019 F. Sferra, M. Krapp, N. Roming, M. Schaeffer, **A. Malik**, B. Hare, R. Brecha. "Towards optimal 1.5°C and 2°C emission pathways for individual countries: A Finland case study". *Energy Policy*.

Reports/Book-chapters

Oct 2026 R. Dikshit, **A. Malik**, V. Chaturvedi. "Stranded costs and equity in net-zero energy transitions in Asia". In *Finance and Energy Transition in Asia*, pp. 30–64. Routledge. doi:10.4324/9781042018000-3

Aug 2026 **A. Malik**, V. Chaturvedi, R. Varma. "Transition Risk Scenarios for India's Financial Sector: Adapting Global Frameworks to National Realities and Financial Stability Risks". *CEEW Issue Brief*.

Dec 2025 CEEW. "Assessment of net zero strategies for Uttarakhand" (Under review).

Nov 2025 CEEW, Ahmedabad University, and UNOPS. "Roadmap for a net zero power sector in Gujarat".

Apr 2025 CEEW, Ahmedabad University, and UNOPS. "Assessment of net zero scenarios for Gujarat".

Dec 2024 P Das, ZA Khan, ... A. Malik. *"Supporting climate policy for Indian states: Learnings from long-term modelling". CEEW Issue Brief*

Policy Blogs

2026 **A. Malik**, S. Elango. "Why India must plan for AI data centres' energy and water footprint". *CEEW Blog*.

2023 **A. Malik**, N. Shukla. "How Energy System Modelling Tools Can Inform Design of India's Carbon Emission Trading Scheme". *CEEW Blog*.

Op-eds & Media

2024 **A. Malik**, V. Chaturvedi. "4 Factors to Influence Our Emissions Trading System". *New Indian Express*.

2024 **A. Malik**, P. Das. "Navigating India's Decarbonisation Challenges". *Moneycontrol*.

Positions, Awards, Honors and Recognitions

2024–2026 **Part of CEEW's team**, *Expert Committee on Reserve Bank's Climate-related Information System*

2023 **IOP Trusted Reviewer**, *In recognition of exceptionally high level of peer review consistency*, IOP Publishing

12/2019–03/2020 **Guest exchange scholarship**, *COMMIT project*, PBL, Netherlands
Fully funded 3-month long exchange at TERI, New Delhi, India

04–09/2016 **Merit scholarship worth 3250€**, *University of Hamburg, Germany*

06–08/2013 **Summer Fellowship**, *Jawaharlal Nehru Center for Advanced Scientific Research*, Bangalore, India

2012–2013 **Bursary of £1250**, *University of Oxford, Department of Continuing Education*
Scholarship for online course titled 'Constructing and Analyzing High Resolution Climate Scenarios'

02/2011 **Expedition Grant worth US$ 8500**, *for the 'Students on Ice' Antarctic University Expedition (uantarctic.org)*
Grants provided by Indraprastha University, Delhi and National Center for Antarctic and Ocean Research, Goa

2009 **Certificate of Merit**, *Central Board of Secondary Education*
For outstanding academic performance and for being among the 0.1% of all successful candidates in Physics in the All India Secondary School Examination

2009 **Full Scholarship**, *GIV Engineering Program*, University of Vermont
Fully funded scholarship to attend an engineering summer school

Review work

Reviewer for the following journals:

Nature
Environmental Research Letters: Climate
Climate Policy
Energy Policy
Climate Economics
Energy Research and Social Science
Environmental Research Letters

Computer Skills

proficient	MS Office, R
good	Python, QGIS
basic	GAMS

Languages

Hindi	**native**
English	**near-native fluency**
German	**very good, level C1**

Talks, Lecture, Panel discussions

04/2026 **Guest lecture**, *Economics and Geopolitics of Energy*, WWF-India and OP Jindal University, Students of LL.M. ELECC 2025

02/2026 **Invited talk**, *The Emerging Carbon Markets Landscape in India*, Centre for Nature-based Climate Solutions, National University of Singapore

12/2025 **Panelist**, *Session on carbon removal in Article 6*, CO2 Forward Carbon Removal Summit, Carbon Removal Alliance of India
Delhi

10/2025 **Guest lecture**, *Economics of Climate Change*, WWF-India and OP Jindal University, Students of LL.M. ELECC 2025-2026

10/2025 **Moderator**, *Panel discussion on Decarbonisation of Economy – Leveraging Carbon Credits*, Viksit Gujarat Regional Summit, Ganpat University
Mehsana

08/2025 **Guest lecture**, *Energy Transition and Decarbonisation*, IOCL Senior Management Programme with IIM Calcutta

07/2025 **Speaker**, *India Emissions and Net Zero*, Indian Energy Forum, New Delhi

05/2025 **Guest lecture**, *Decision maker under uncertainty: Insights from modelling*, National Thermal Power Corporation

03/2025 **Panelist**, *Panelist at Sustainable Futures Collaborative's Tufts Climate Panel Discussion*, New Delhi

01/2025 **Guest lecture**, *Introduction to Modelling*, IIT Roorkee

09/2024 **Speaker**, *Conference on Joint Response to Climate Crisis*, Seoul, South Korea
Sookmyung Women's University

09/2024 **Speaker**, *Lecture on Climate Change at the Rajya Sabha (Upper House) for Parliamentary Interns*, New Delhi, India
To interns of Members of Parliament

09/2024 **Speaker**, *IIT Delhi workshop on Energy and Well-being – Indian Case Study*, New Delhi, India
IIT Delhi

09/2024 **Speaker**, *Surya Foundation Think Tank Strengthening Some Key Areas for Viksit Bharat 2047*, New Delhi

08/2024 **Speaker and Panelist**, *Carbon Markets in India*, New Delhi, India
Centre for Social and Economic Progress

08/2024 **Speaker**, *State Dialogue on Accelerating Energy Efficiency in Gujarat*, Vadodara, India
[Watch Video]

02/2024 **Guest speaker**, *Webinar on Role of ETS in India's Net-Zero targets*, Delhi, India
Organised by the Carbon Markets Association of India

02/2024 **Expert speaker**, *Accelerating the Just Transition of MSMEs: Capacity Development of Policymakers Program*, Lucknow, India
Organised by School of Public Policy and UNDP

01/2024 **Panelist**, *Impact of Net-Zero Transition Policies on Clean Jobs, Skilling Needs and Inter-state Migration*, New Delhi, India
Event organised by ICRIER — [Watch Video]

01/2024 **Speaker**, *Implications of India's ETS on its net-zero target*, New Delhi, India
Niti Aayog

11/2023 **Oral presentation**, *Integrated Assessment Modelling Consortium*, Venice, Italy
Role of ETS in India's Net-zero targets: A modelling-based assessment. Presented with a fully-funded scholarship from EU DG CLIMA

03/2022 **Defense presentation**, *Barriers to power sector decarbonisation in India*, TU Berlin, Germany

12/2020 **Oral presentation**, *Integrated Assessment Modelling Consortium*, Online
Assumptions on employment factor approach in energy scenarios

Aman Malik and Christoph Bertram

09/2020 **Expert Panel**, *Climate Action and Renewable Energy in context of a Green Recovery*, Africa–EU Climate Partnership, Online

07/2019 **Presentation**, *Preventing stranded assets and empowering renewables*, CD-LINKS Summer School, Venice
Italy. Fully-funded participation (06–07/2019) in a week-long summer school on Integrated Assessment Models

11/2018 **Poster presentation**, *Integrated Assessment Modelling Consortium*, Seville, Spain
'Not (yet) another coal story: Power-sector lock-ins in India - perspective from national and global models'
Aman Malik and Christoph Bertram